

Computer Vision HW#1

Nima Darbandi

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 nimadarbandi / ComputerVision 



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 20 lines (14 loc) · 469 Bytes

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```
1  import numpy as np
2  import matplotlib.pyplot as plt
3
4  input_file = 'test3.img'
5  output_file = 'test3B.img'
6  # Read the raw binary data
7  image_data = np.fromfile(input_file, dtype=np.uint8)
8
9
10 # Reshape the image data into a 512x512 array
11 image_data = image_data[512:]
12 image_data = image_data.reshape((512, 512))
13
14 #Thresholding
15 threshold_value = 128
16 binary_image = np.where(image_data > threshold_value, 255, 0).astype(np.uint8)
17
18 #save image
19 binary_image.tofile(output_file)
```

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Code

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273 lines (236 loc) · 11.1 KB

Raw

```
1  import numpy as np
2  import matplotlib.pyplot as plt
3  import pprint
4  from math import sqrt
5
6  # Inputs
7  input_file = 'test3B.img' #the binary image B
8  min_size = 500 #maximum pixel count for a component to be considered valid
9  foreground = 255 #the value of the background pixel
10 # Outputs
11 area = {} #the component size for each component
12 centroid = {} # the location of the centroid of each component
13 bounding_box = {} #the coordinates of the bounding box for each component
14 theta = {} #the orientation of axis of ellongation of each component
15 eccentricity = {} #the eccentricity of each component
16 perimeters = {} #the perimeter of each component
17 compactness = {} #the compactness of each component
18 output_file = 'test3C.jpg' #the color coded display of the components of the image B
19
20 #####Functions#####
21 #####
22
23 def iterative_CCL(binary_image):
24     height, width = 512, 512
25     labels = np.zeros_like(binary_image, dtype=int)
26     #print(labels.shape)
27
28     new_label = 1
29     # equivalence table
30     equivalence = {}
31
32
33     for i in range(height):
34         for j in range(width):
35             #print(binary_image[i, j],end=",")
36             if binary_image[i, j] == foreground: # If pixel is part of a component
37
38                 neighbor_labels = [0, 0] #upper, left pixels
39                 if i > 0 and labels[i-1, j] > 0: #upper pixel has label
40                     neighbor_labels[0] = labels[i-1, j]
41                 if j > 0 and labels[i, j-1] > 0: #left pixel has label
42                     neighbor_labels[1] = labels[i, j-1]
43
44                 if neighbor_labels[0] > 0 or neighbor_labels[1] > 0:
45                     if neighbor_labels[0] == 0 : #only left pixel has label
46                         labels[i, j] = neighbor_labels[1]
47                     elif neighbor_labels[1] == 0 : #only upper pixel has label
48                         labels[i, j] = neighbor_labels[0]
49                     elif neighbor_labels[0] == neighbor_labels[1] : #both upper and left pixels have the same label
50                         labels[i, j] = neighbor_labels[0]
51                     elif neighbor_labels[0] != neighbor_labels[1] : #both upper and left pixels have different labels
52                         labels[i, j] = neighbor_labels[0]
53                     # Record equivalence: all neighbor labels become equivalent.
54                     if neighbor_labels[0] in equivalence:
55                         equivalence[neighbor_labels[0]].add(int(neighbor_labels[1]))
56                     else:
57                         equivalence[neighbor_labels[0]] = {int(neighbor_labels[1])}
58                     if neighbor_labels[1] in equivalence:
59                         equivalence[neighbor_labels[1]].add(int(neighbor_labels[0]))
60                     else:
61                         equivalence[neighbor_labels[1]] = {int(neighbor_labels[0])}
```

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```
62         else:
63             labels[i, j] = new_label
64             equivalence[new_label] = {new_label}
65             new_label += 1
66     #pprint.pprint(equivalence)
67     #print(equivalence.items())
68
69     # merge overlapping sets in equivalence table until stable.
70     changed = True
71     while changed:
72         changed = False
73         for key in list(equivalence.keys()):
74             merged_set = set(equivalence[key])
75             for lbl in list(equivalence[key]):
76                 merged_set = merged_set.union(equivalence.get(lbl, set()))
77             if merged_set != equivalence[key]:
78                 equivalence[key] = merged_set
79                 changed = True
80
81     # Build a final mapping table based on the equivalence table (for each label, assign the minimum label from its equ
82     final_map = {}
83     for key, eq_set in equivalence.items():
84         min_label = min(eq_set)
85         for l in eq_set:
86             final_map[l] = min_label
87
88     #pprint.pprint(final_map)
89
90     # Replace each label in the image using final_map.
91     for i in range(height):
92         for j in range(width):
93             if labels[i, j] > 0:
94                 labels[i, j] = final_map[labels[i, j]]
95
96     return labels
97
98     #####
99
100     def count_components(labels, min_size):
101         valid_components = 0
102         area = {}
103         centroid = {}
104         bounding_box = {}
105         theta = {}
106         eccentricity = {}
107         perimeters = {}
108         compactness = {}
109
110         # Count pixels for each label
111         unique_labels, counts = np.unique(labels[labels > 0], return_counts=True)
112         for i in range(len(unique_labels)):
113             print(f"Component {unique_labels[i]}: {counts[i]}")
114             if counts[i] >= min_size:
115                 valid_components += 1
116                 #area
117                 area[unique_labels[i]] = counts[i]
118                 #component coordinates
119                 row, col = np.where(labels == unique_labels[i])
120                 #centroid
121                 centroid_row = np.mean(row)
122                 centroid_col = np.mean(col)
123                 centroid[unique_labels[i]] = (centroid_row, centroid_col)
124                 #bounding box
125                 min_row = np.min(row)
126                 min_col = np.min(col)
127                 max_row = np.max(row)
128                 max_col = np.max(col)
129                 bounding_box[unique_labels[i]] = (min_row, min_col, max_row, max_col)
130                 #print("bounding box", unique_labels[i], bounding_box[unique_labels[i]])
131                 #orientation
132                 a = np.sum((row - centroid_row) ** 2)
133                 b = np.sum((col - centroid_col) ** 2)
134                 c = np.sum((row - centroid_row) * (col - centroid_col))
135                 theta[unique_labels[i]] = 0.5 * np.arctan2(2 * c, a - b)
136                 #eccentricity
137                 #print("starting eccentricity", unique_labels[i])
138                 delta = 0.5 * np.sqrt((a - b) ** 2 + 4 * c ** 2)
```

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```
137         #print("delta", unique_labels[i], delta)
138         E1 = (a + b + delta) / 2 #largest eigenvalue
139         E2 = (a + b - delta) / 2 #smallest eigenvalue
140         ER = E2 / E1
141         if E1 > 0:
142             eccentricity[unique_labels[i]] = np.sqrt(1 - (E2 / E1)**2)
143         else:
144             eccentricity[unique_labels[i]] = 0
145         #print("starting perimeter", unique_labels[i])
146         #perimeter
147         component_label = (labels == unique_labels[i]).astype(np.uint8)
148         boundary = trace_boundary(component_label)
149         #print("boundary", unique_labels[i], boundary)
150         perim = 0
151         for j in range(1, len(boundary)):
152             r1, c1 = boundary[j-1]
153             r2, c2 = boundary[j]
154             dr = abs(r2 - r1)
155             dc = abs(c2 - c1)
156             if dr == 1 and dc == 1:
157                 perim += sqrt(2)
158             else:
159                 perim += 1
160         perimeters[unique_labels[i]] = perim
161         #print("starting compactness", unique_labels[i])
162         #compactness
163         compactness[unique_labels[i]] = (perimeters[unique_labels[i]] ** 2) / area[unique_labels[i]]
164         #print("finished component", unique_labels[i])
165
166     return valid_components, area, centroid, bounding_box, theta, eccentricity, perimeters, compactness
167
168     #####
169
170     def trace_boundary(component_label):
171
172         # getting component pixels
173         rows, cols = np.where(component_label == 1)
174         if len(rows) == 0:
175             return [] # no component pixels
176         s = (rows[0], cols[0]) # starting pixel for the boundary
177         boundary_list = [s]
178
179         # previous pixel = the 4-neighbor to the west of s.
180         p = (s[0], s[1] - 1)
181
182         # Define 8-neighbor in clockwise order starting from West.
183         eight_neighbor = [(0, -1), (-1, -1), (-1, 0), (-1, 1),
184                           (0, 1), (1, 1), (1, 0), (1, -1)]
185
186
187         try:
188             start_pixel = eight_neighbor.index((0, -1))
189         except ValueError:
190             start_pixel = 0
191
192         current = s
193         visited = set()
194         visited.add(current)
195
196         while True:
197             found = False
198             next_pixel = None
199             # Examine the 8 neighbors of 'current' in clockwise order starting at start_pixel.
200             for i in range(8):
201                 idx = (start_pixel + i) % 8
202                 offset = eight_neighbor[idx]
203                 candidate = (current[0] + offset[0], current[1] + offset[1])
204                 # Ensure candidate is in image pixels.
205                 if (candidate[0] < 0 or candidate[0] >= component_label.shape[0] or
206                     candidate[1] < 0 or candidate[1] >= component_label.shape[1]):
207                     continue
208                 if component_label[candidate] == 1 and candidate not in visited:
209                     # The first neighbor that is part of the component.
210                     next_pixel = candidate
211                     found = True
```

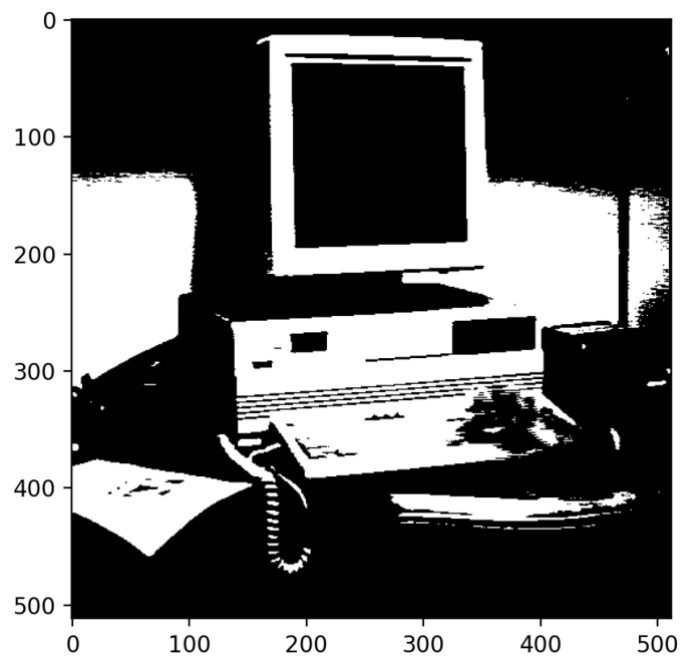
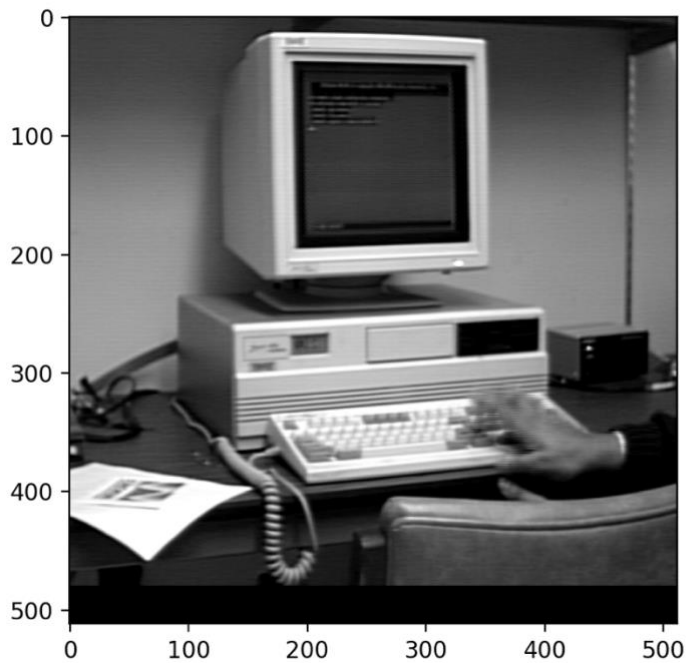
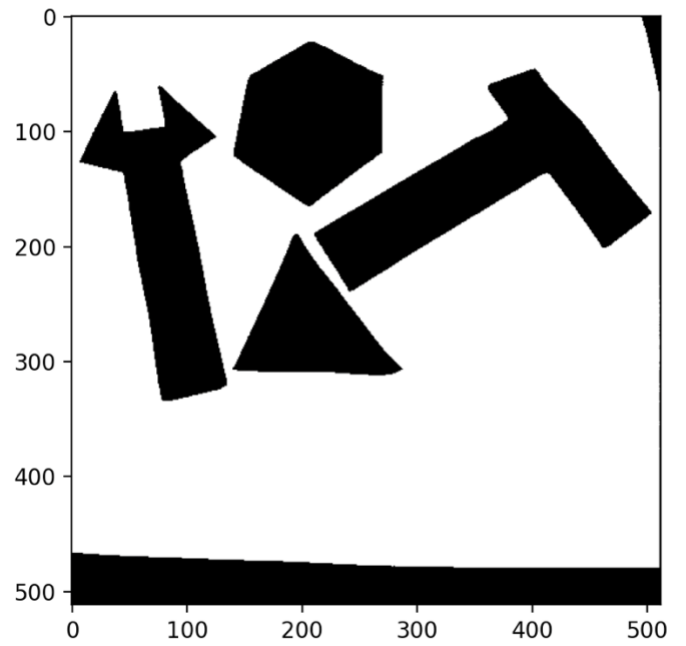
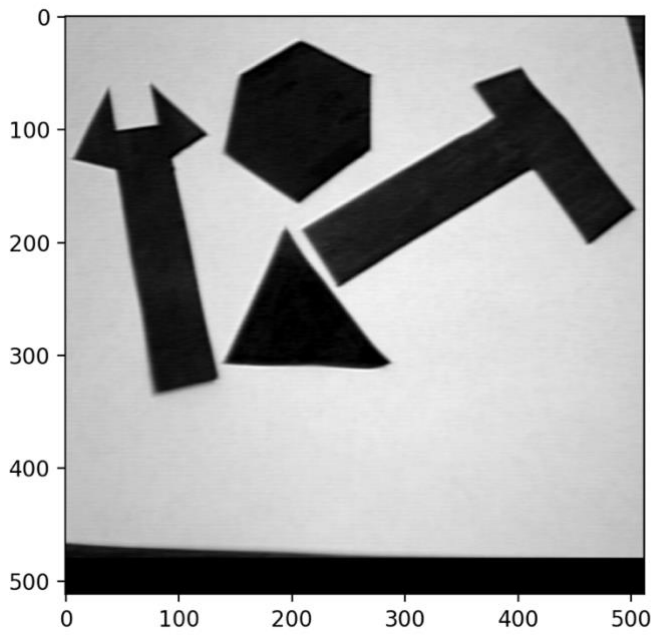
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```
212         # Set the new start
213         start_pixel = (idx - 1) % 8
214         break
215     if not found:
216         # If no neighbor is found
217         break
218     # Add the found boundary pixel.
219     boundary_list.append(next_pixel)
220     visited.add(next_pixel)
221     if next_pixel == s:
222         break
223     current = next_pixel
224
225     return boundary_list
226
227     #####
228     def Paint_image(labels, output_file, min_size):
229         # Create an RGB image
230         rgb_image = np.zeros((labels.shape[0], labels.shape[1], 3), dtype=np.uint8)
231
232         # Get unique labels
233         unique_labels, counts = np.unique(labels[labels > 0], return_counts=True)
234
235         # random colors for each label
236         for j in range(len(unique_labels)):
237             if counts[j] >= min_size:
238                 color = np.random.randint(0, 256, size=3)
239                 rgb_image[labels == unique_labels[j]] = color
240
241         # Save the RGB image
242         plt.imshow(output_file, rgb_image)
243
244     #####Main#####
245     #####
246
247     # Read the binary image:
248     binary_image = np.fromfile(input_file, dtype=np.uint8)
249     #header stripping
250     binary_image = binary_image[512:]
251     binary_image = binary_image.reshape((512, 512)) # adjust dimensions as needed
252
253     #connected component labeling
254     labels = iterative_CCL(binary_image)
255
256     print("count started")
257     total_components, area, centroid, bounding_box, theta, eccentricity, perimeters, compactness = count_components(labels,
258     print("count finished")
259
260     for label in area:
261         print(f"Component {label}:")
262         print(f" Area: {area[label]}")
263         print(f" Centroid: ({centroid[label][0]:.2f}, {centroid[label][1]:.2f})")
264         print(f" Bounding Box(min to max corners): min:({bounding_box[label][0]:.0f}, {bounding_box[label][1]:.0f}), max:({
265         print(f" Orientation (Theta): {theta[label]:.2f} radians")
266         print(f" Eccentricity: {eccentricity[label]:.2f}")
267         print(f" Perimeter: {perimeters[label]:.2f}")
268         print(f" Compactness: {compactness[label]:.2f}")
269
270     print(f"\nTotal number of components: {total_components}\n")
271
272
273     Paint_image(labels, output_file, min_size)
```

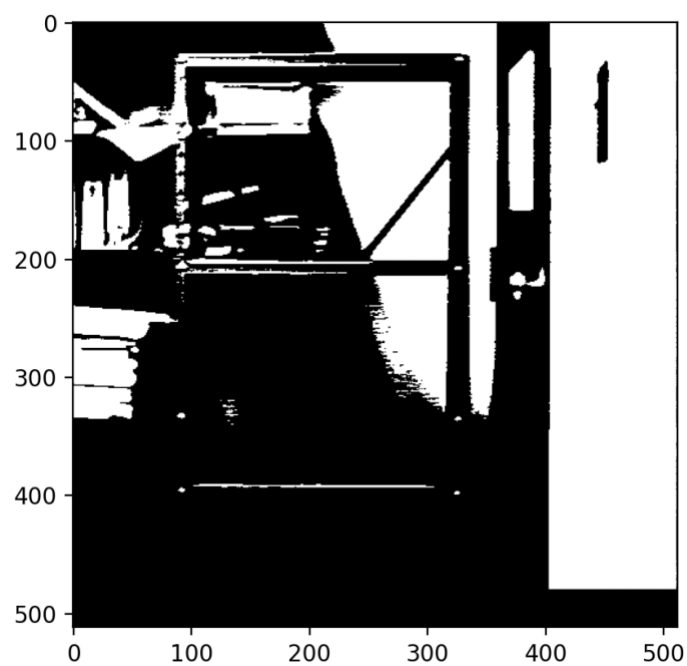
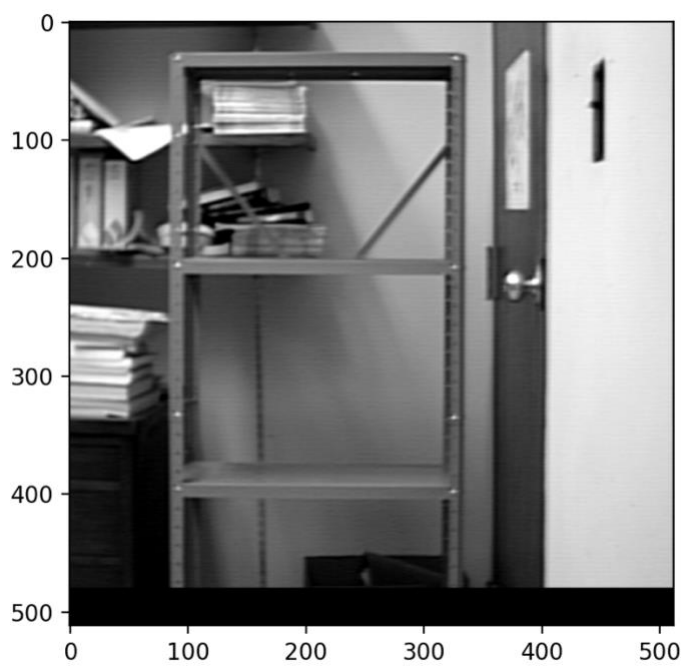
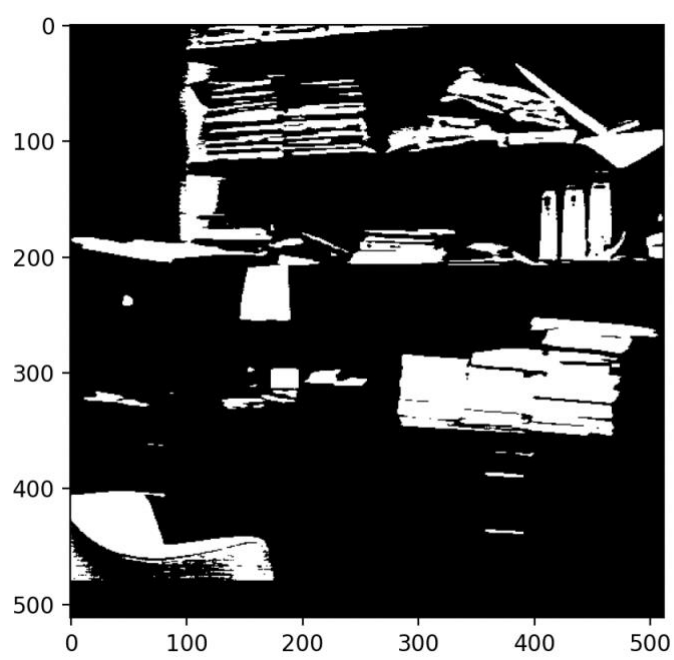
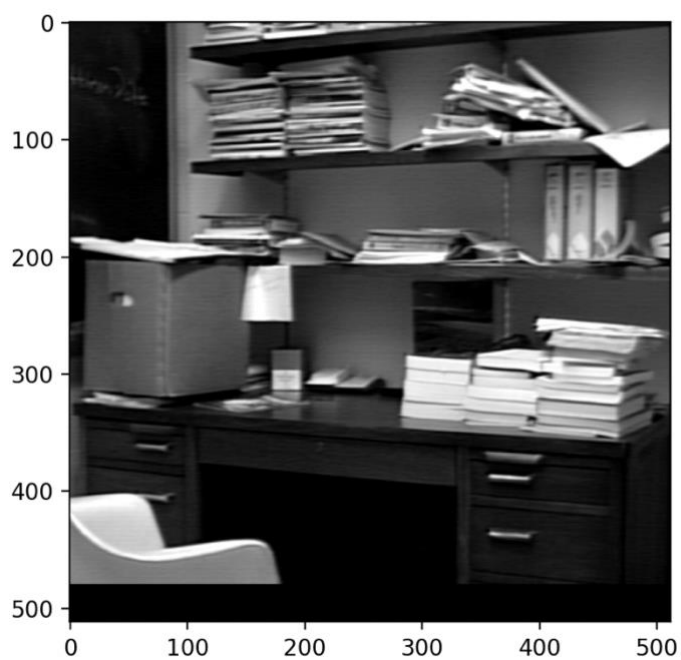
2. Hard copies of images B and B_T



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3,4. Hardcopies of the images C, for three values of the minimum size specification (i.e., size filter parameter) along with a description of the components in C. Comments on the results obtained in each case. In particular, comment on the tradeoff involved in the selection of the size filter parameter.

I passed four different binary images `comb.img`, `test1B.img`, `test2B.img`, `test3B.img` with a minimum component size of 200, 500, 1000 .

As I increased the `min_size`, the number of total components reported decreased as we expected, and vice versa. If we want more detailed components it is good to select smaller `min_size` however, selecting very small `min_size` (say 50pixels or lower) led to selecting pixel errors around the image, which is considered as noise and we are not interested in. so I selected 200, 500, 1000 to ignore the background noises as much as I can.

I also specified the pixel of interest or foreground. Foreground selection was important because for each image the objects were coded differently in terms of gray levels and it led to different values for background in binary images. For example in `comb.img` the background was white(255) but for the test images the background was black(0) after thresholding.

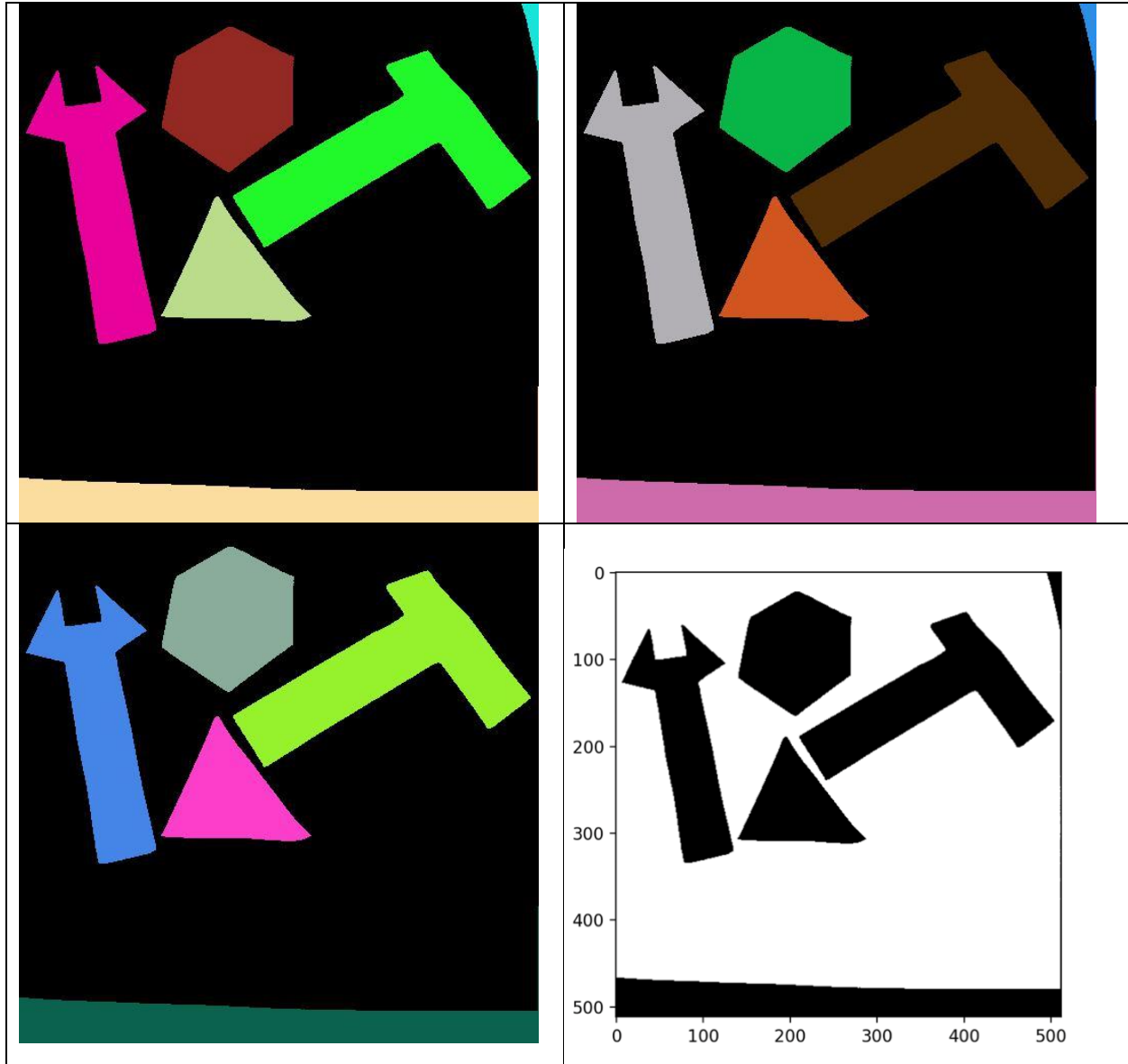
So I put a parameter to specify the foreground(pixel value of interest) in the code. In terms of minimum pixels size.

From the next page you can see the program inputs and outputs for each of these for image files:

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From top left 200, 500, 1000 pixels min size and 6. 6, 5 components are detected.
Here is the output data respectively:

200:

Component 1:

Area: 620

Centroid: (28.17, 506.45)

Bounding Box(min to max corners): min:(0, 495), max:(114, 511)

Orientation (Theta): 0.09 radians

Eccentricity: 0.94

Perimeter: 130.00

Compactness: 27.26

Component 2:

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Area: 13148
Centroid: (91.73, 207.34)
Bounding Box(min to max corners): min:(23, 141), max:(165, 270)
Orientation (Theta): -0.71 radians
Eccentricity: 0.39
Perimeter: 15230.23
Compactness: 17642.23
Component 26:
Area: 19762
Centroid: (146.56, 361.07)
Bounding Box(min to max corners): min:(46, 211), max:(239, 503)
Orientation (Theta): -1.23 radians
Eccentricity: 0.87
Perimeter: 22276.25
Compactness: 25110.38
Component 50:
Area: 15327
Centroid: (196.63, 81.29)
Bounding Box(min to max corners): min:(61, 7), max:(334, 134)
Orientation (Theta): 0.19 radians
Eccentricity: 0.92
Perimeter: 1630.80
Compactness: 173.52
Component 199:
Area: 8913
Centroid: (269.84, 207.19)
Bounding Box(min to max corners): min:(190, 140), max:(312, 287)
Orientation (Theta): 0.96 radians
Eccentricity: 0.57
Perimeter: 4428.36
Compactness: 2200.20
Component 278:
Area: 18472
Centroid: (492.97, 241.07)
Bounding Box(min to max corners): min:(377, 0), max:(511, 511)
Orientation (Theta): 1.56 radians
Eccentricity: 0.94
Perimeter: 16784.93
Compactness: 15251.93

Total number of components: 6

500:

Component 1:
Area: 620
Centroid: (28.17, 506.45)
Bounding Box(min to max corners): min:(0, 495), max:(114, 511)
Orientation (Theta): 0.09 radians
Eccentricity: 0.94
Perimeter: 130.00
Compactness: 27.26
Component 2:
Area: 13148
Centroid: (91.73, 207.34)
Bounding Box(min to max corners): min:(23, 141), max:(165, 270)
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Area: 18472
Centroid: (492.97, 241.07)
Bounding Box(min to max corners): min:(377, 0), max:(511, 511)
Orientation (Theta): 1.56 radians
Eccentricity: 0.94
Perimeter: 16784.93
Compactness: 15251.93

Total number of components: 6

1000:

Component 2:

Area: 13148
Centroid: (91.73, 207.34)
Bounding Box(min to max corners): min:(23, 141), max:(165, 270)
Orientation (Theta): -0.71 radians
Eccentricity: 0.39
Perimeter: 15230.23
Compactness: 17642.23

Component 26:

Area: 19762
Centroid: (146.56, 361.07)
Bounding Box(min to max corners): min:(46, 211), max:(239, 503)
Orientation (Theta): -1.23 radians
Eccentricity: 0.87
Perimeter: 22276.25
Compactness: 25110.38

Component 50:

Area: 15327
Centroid: (196.63, 81.29)
Bounding Box(min to max corners): min:(61, 7), max:(334, 134)
Orientation (Theta): 0.19 radians
Eccentricity: 0.92
Perimeter: 1630.80
Compactness: 173.52

Component 199:

Area: 8913
Centroid: (269.84, 207.19)
Bounding Box(min to max corners): min:(190, 140), max:(312, 287)
Orientation (Theta): 0.96 radians
Eccentricity: 0.57
Perimeter: 4428.36
Compactness: 2200.20

Component 278:

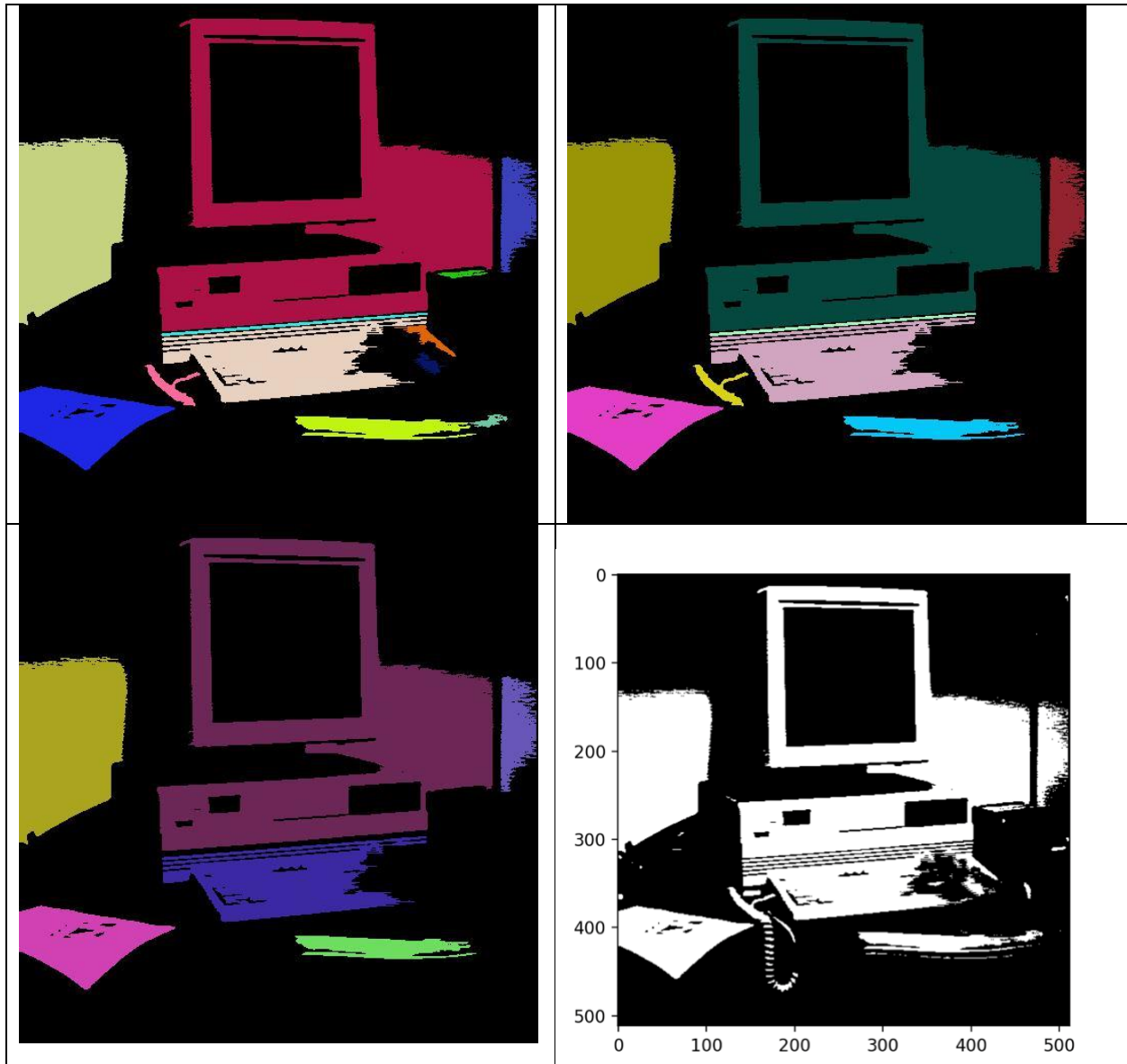
Area: 18472
Centroid: (492.97, 241.07)
Bounding Box(min to max corners): min:(377, 0), max:(511, 511)
Orientation (Theta): 1.56 radians
Eccentricity: 0.94
Perimeter: 16784.93
Compactness: 15251.93

Total number of components: 5

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From top left 200, 500, 1000 pixels min size and 12, 8, 6 components are detected.
Here is the output data respectively:

200:

Component 1:

Area: 39759

Centroid: (200.58, 305.78)

Bounding Box(min to max corners): min:(14, 136), max:(321, 467)

Orientation (Theta): -1.55 radians

Eccentricity: 0.45

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Perimeter: 392.57
Compactness: 3.88

Component 25:

Area: 15669
Centroid: (211.22, 47.66)
Bounding Box(min to max corners): min:(131, 0), max:(315, 106)
Orientation (Theta): -0.22 radians
Eccentricity: 0.79
Perimeter: 37.90
Compactness: 0.09

Component 96:

Area: 2771
Centroid: (201.01, 489.50)
Bounding Box(min to max corners): min:(150, 475), max:(263, 509)
Orientation (Theta): -0.06 radians
Eccentricity: 0.90
Perimeter: 15.66
Compactness: 0.09

Component 213:

Area: 200
Centroid: (265.12, 436.63)
Bounding Box(min to max corners): min:(262, 413), max:(269, 461)
Orientation (Theta): -1.51 radians
Eccentricity: 0.94
Perimeter: 15.00
Compactness: 1.12

Component 248:

Area: 719
Centroid: (315.30, 265.57)
Bounding Box(min to max corners): min:(304, 140), max:(326, 402)
Orientation (Theta): -1.49 radians
Eccentricity: 0.94
Perimeter: 543.40
Compactness: 410.68

Component 253:

Area: 11077
Centroid: (351.15, 262.32)
Bounding Box(min to max corners): min:(308, 141), max:(392, 401)
Orientation (Theta): -1.54 radians
Eccentricity: 0.90
Perimeter: 116.38
Compactness: 1.22

Component 279:

Area: 406
Centroid: (327.84, 402.55)
Bounding Box(min to max corners): min:(316, 370), max:(346, 430)
Orientation (Theta): 1.03 radians
Eccentricity: 0.93
Perimeter: 97.98
Compactness: 23.65

Component 373:

Area: 270
Centroid: (353.93, 403.27)
Bounding Box(min to max corners): min:(343, 389), max:(364, 419)
Orientation (Theta): 0.97 radians
Eccentricity: 0.85
Perimeter: 67.28
Compactness: 16.77

Component 388:

Area: 593
Centroid: (375.89, 150.43)
Bounding Box(min to max corners): min:(355, 124), max:(395, 178)
Orientation (Theta): 1.01 radians
Eccentricity: 0.85
Perimeter: 207.55
Compactness: 72.64

Component 438:

Area: 6680
Centroid: (410.91, 60.59)

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Bounding Box(min to max corners): min:(376, 0), max:(458, 153)
Orientation (Theta): 1.52 radians
Eccentricity: 0.84
Perimeter: 537.82
Compactness: 43.30
Component 459:
Area: 216
Centroid: (410.26, 464.25)
Bounding Box(min to max corners): min:(404, 444), max:(416, 481)
Orientation (Theta): -1.33 radians
Eccentricity: 0.92
Perimeter: 63.11
Compactness: 18.44
Component 467:
Area: 2668
Centroid: (416.88, 354.15)
Bounding Box(min to max corners): min:(406, 272), max:(428, 450)
Orientation (Theta): 1.54 radians
Eccentricity: 0.94
Perimeter: 335.33
Compactness: 42.15

Total number of components: 12

500:

Component 1:
Area: 39759
Centroid: (200.58, 305.78)
Bounding Box(min to max corners): min:(14, 136), max:(321, 467)
Orientation (Theta): -1.55 radians
Eccentricity: 0.45
Perimeter: 392.57
Compactness: 3.88
Component 25:
Area: 15669
Centroid: (211.22, 47.66)
Bounding Box(min to max corners): min:(131, 0), max:(315, 106)
Orientation (Theta): -0.22 radians
Eccentricity: 0.79
Perimeter: 37.90
Compactness: 0.09
Component 96:
Area: 2771
Centroid: (201.01, 489.50)
Bounding Box(min to max corners): min:(150, 475), max:(263, 509)
Orientation (Theta): -0.06 radians
Eccentricity: 0.90
Perimeter: 15.66
Compactness: 0.09
Component 248:
Area: 719
Centroid: (315.30, 265.57)
Bounding Box(min to max corners): min:(304, 140), max:(326, 402)
Orientation (Theta): -1.49 radians
Eccentricity: 0.94
Perimeter: 543.40
Compactness: 410.68
Component 253:
Area: 11077
Centroid: (351.15, 262.32)
Bounding Box(min to max corners): min:(308, 141), max:(392, 401)
Orientation (Theta): -1.54 radians
Eccentricity: 0.90
Perimeter: 116.38
Compactness: 1.22
Component 388:
Area: 593
Centroid: (375.89, 150.43)
Bounding Box(min to max corners): min:(355, 124), max:(395, 178)

Computer Vision HW#1

Nima Darbandi

811296007

Orientation (Theta): 1.01 radians
Eccentricity: 0.85
Perimeter: 207.55
Compactness: 72.64
Component 438:
Area: 6680
Centroid: (410.91, 60.59)
Bounding Box(min to max corners): min:(376, 0), max:(458, 153)
Orientation (Theta): 1.52 radians
Eccentricity: 0.84
Perimeter: 537.82
Compactness: 43.30
Component 467:
Area: 2668
Centroid: (416.88, 354.15)
Bounding Box(min to max corners): min:(406, 272), max:(428, 450)
Orientation (Theta): 1.54 radians
Eccentricity: 0.94
Perimeter: 335.33
Compactness: 42.15

Total number of components: 8

1000:

Component 1:
Area: 39759
Centroid: (200.58, 305.78)
Bounding Box(min to max corners): min:(14, 136), max:(321, 467)
Orientation (Theta): -1.55 radians
Eccentricity: 0.45
Perimeter: 392.57
Compactness: 3.88
Component 25:
Area: 15669
Centroid: (211.22, 47.66)
Bounding Box(min to max corners): min:(131, 0), max:(315, 106)
Orientation (Theta): -0.22 radians
Eccentricity: 0.79
Perimeter: 37.90
Compactness: 0.09
Component 96:
Area: 2771
Centroid: (201.01, 489.50)
Bounding Box(min to max corners): min:(150, 475), max:(263, 509)
Orientation (Theta): -0.06 radians
Eccentricity: 0.90
Perimeter: 15.66
Compactness: 0.09
Component 253:
Area: 11077
Centroid: (351.15, 262.32)
Bounding Box(min to max corners): min:(308, 141), max:(392, 401)
Orientation (Theta): -1.54 radians
Eccentricity: 0.90
Perimeter: 116.38
Compactness: 1.22
Component 438:
Area: 6680
Centroid: (410.91, 60.59)
Bounding Box(min to max corners): min:(376, 0), max:(458, 153)
Orientation (Theta): 1.52 radians
Eccentricity: 0.84
Perimeter: 537.82
Compactness: 43.30
Component 467:
Area: 2668
Centroid: (416.88, 354.15)
Bounding Box(min to max corners): min:(406, 272), max:(428, 450)

Computer Vision HW#1

Nima Darbandi

811296007

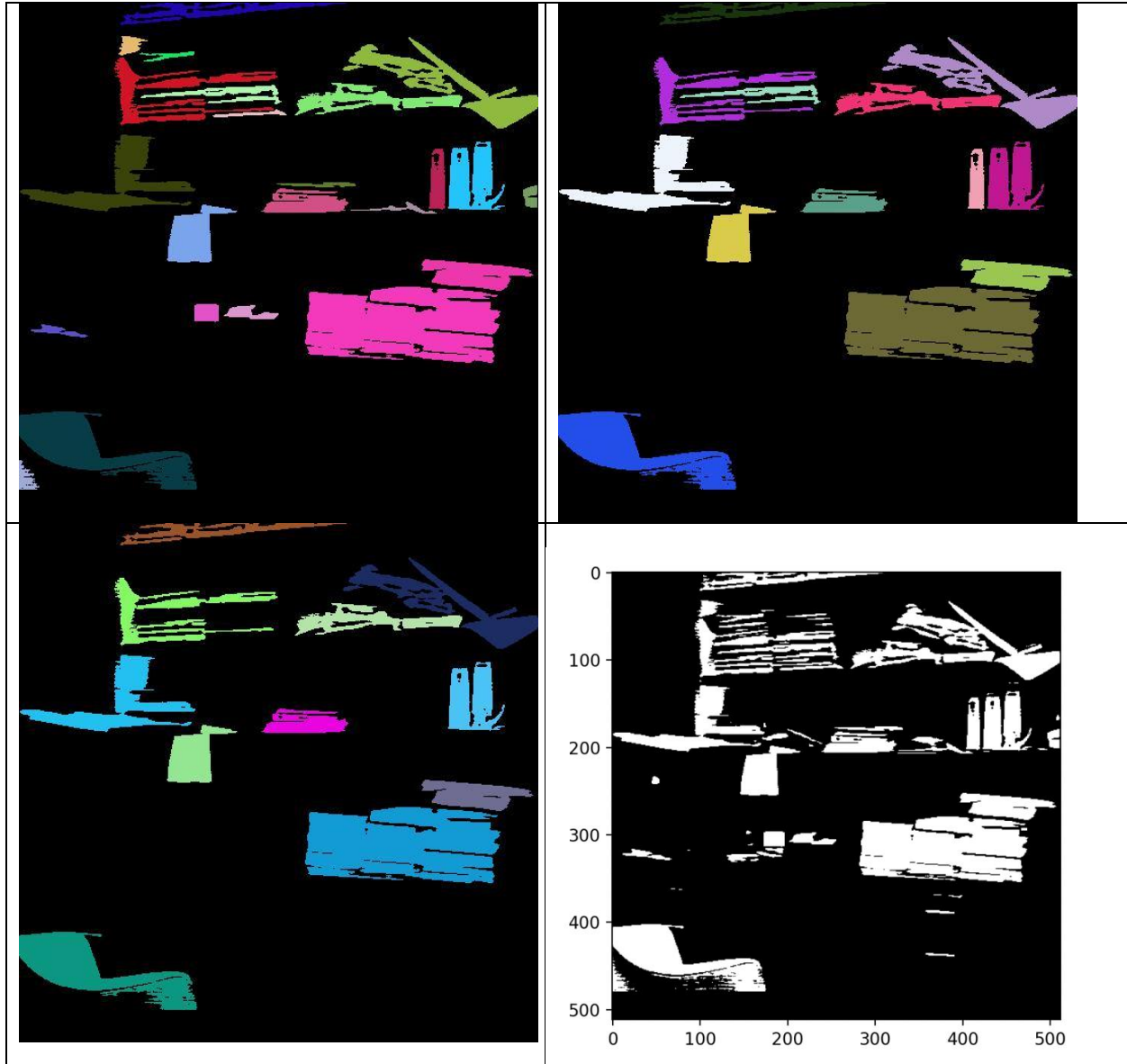
Orientation (Theta): 1.54 radians

Eccentricity: 0.94

Perimeter: 335.33

Compactness: 42.15

Total number of components: 6



From top left 200, 500, 1000 pixels min size and 23, 13, 11 components are detected.
Here is the output data respectively:

200:

Component 1:

Area: 1614

Centroid: (7.35, 188.34)

Bounding Box(min to max corners): min:(0, 100), max:(21, 322)

Orientation (Theta): -1.50 radians

Eccentricity: 0.94

Perimeter: 197.11

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Compactness: 24.07

Component 47:

Area: 305

Centroid: (41.29, 109.33)

Bounding Box(min to max corners): min:(33, 98), max:(51, 126)

Orientation (Theta): -1.10 radians

Eccentricity: 0.69

Perimeter: 26.66

Compactness: 2.33

Component 48:

Area: 3997

Centroid: (80.32, 424.61)

Bounding Box(min to max corners): min:(34, 319), max:(123, 511)

Orientation (Theta): 1.21 radians

Eccentricity: 0.93

Perimeter: 562.67

Compactness: 79.21

Component 63:

Area: 205

Centroid: (51.83, 147.06)

Bounding Box(min to max corners): min:(48, 119), max:(57, 172)

Orientation (Theta): -1.48 radians

Eccentricity: 0.94

Perimeter: 58.90

Compactness: 16.92

Component 80:

Area: 2847

Centroid: (88.75, 151.81)

Bounding Box(min to max corners): min:(54, 94), max:(120, 254)

Orientation (Theta): -1.51 radians

Eccentricity: 0.89

Perimeter: 91.28

Compactness: 2.93

Component 150:

Area: 2089

Centroid: (95.38, 345.07)

Bounding Box(min to max corners): min:(79, 272), max:(110, 436)

Orientation (Theta): 1.57 radians

Eccentricity: 0.94

Perimeter: 45.24

Compactness: 0.98

Component 157:

Area: 929

Centroid: (89.20, 206.00)

Bounding Box(min to max corners): min:(81, 120), max:(99, 258)

Orientation (Theta): -1.56 radians

Eccentricity: 0.94

Perimeter: 25.73

Compactness: 0.71

Component 257:

Area: 216

Centroid: (109.31, 227.94)

Bounding Box(min to max corners): min:(105, 191), max:(112, 263)

Orientation (Theta): -1.52 radians

Eccentricity: 0.94

Perimeter: 159.46

Compactness: 117.71

Component 285:

Area: 3632

Centroid: (176.30, 97.39)

Bounding Box(min to max corners): min:(130, 1), max:(205, 173)

Orientation (Theta): -1.31 radians

Eccentricity: 0.84

Perimeter: 33.24

Compactness: 0.30

Component 288:

Area: 2386

Centroid: (173.81, 446.45)

Bounding Box(min to max corners): min:(137, 423), max:(204, 478)

Computer Vision HW#1

Nima Darbandi

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Orientation (Theta): -0.03 radians

Eccentricity: 0.67

Perimeter: 115.18

Compactness: 5.56

Component 304:

Area: 737

Centroid: (176.52, 412.44)

Bounding Box(min to max corners): min:(144, 404), max:(203, 419)

Orientation (Theta): -0.03 radians

Eccentricity: 0.92

Perimeter: 304.40

Compactness: 125.72

Component 335:

Area: 246

Centroid: (178.94, 297.50)

Bounding Box(min to max corners): min:(177, 254), max:(181, 331)

Orientation (Theta): -1.57 radians

Eccentricity: 0.94

Perimeter: 113.73

Compactness: 52.58

Component 340:

Area: 241

Centroid: (190.71, 504.15)

Bounding Box(min to max corners): min:(180, 497), max:(201, 510)

Orientation (Theta): 0.04 radians

Eccentricity: 0.77

Perimeter: 209.91

Compactness: 182.83

Component 345:

Area: 1384

Centroid: (196.74, 281.48)

Bounding Box(min to max corners): min:(183, 239), max:(206, 324)

Orientation (Theta): -1.57 radians

Eccentricity: 0.91

Perimeter: 57.24

Compactness: 2.37

Component 395:

Area: 229

Centroid: (202.87, 376.72)

Bounding Box(min to max corners): min:(196, 327), max:(207, 411)

Orientation (Theta): 1.55 radians

Eccentricity: 0.94

Perimeter: 34.38

Compactness: 5.16

Component 403:

Area: 2065

Centroid: (229.68, 171.12)

Bounding Box(min to max corners): min:(199, 146), max:(255, 214)

Orientation (Theta): -0.61 radians

Eccentricity: 0.73

Perimeter: 494.17

Compactness: 118.26

Component 440:

Area: 1986

Centroid: (267.26, 447.10)

Bounding Box(min to max corners): min:(253, 397), max:(283, 506)

Orientation (Theta): 1.50 radians

Eccentricity: 0.92

Perimeter: 108.56

Compactness: 5.93

Component 453:

Area: 11222

Centroid: (316.58, 377.34)

Bounding Box(min to max corners): min:(280, 282), max:(354, 474)

Orientation (Theta): 1.55 radians

Eccentricity: 0.90

Perimeter: 115.80

Compactness: 1.19

Component 468:

Computer Vision HW#1

Nima Darbandi

811296007

Area: 405
Centroid: (305.10, 184.52)
Bounding Box(min to max corners): min:(297, 173), max:(314, 196)
Orientation (Theta): 1.56 radians
Eccentricity: 0.70
Perimeter: 407.73
Compactness: 410.47
Component 470:
Area: 404
Centroid: (305.69, 226.58)
Bounding Box(min to max corners): min:(298, 202), max:(311, 255)
Orientation (Theta): 1.46 radians
Eccentricity: 0.92
Perimeter: 392.68
Compactness: 381.68
Component 494:
Area: 265
Centroid: (323.32, 37.62)
Bounding Box(min to max corners): min:(317, 12), max:(329, 67)
Orientation (Theta): 1.42 radians
Eccentricity: 0.94
Perimeter: 125.87
Compactness: 59.79
Component 529:
Area: 5655
Centroid: (440.19, 76.64)
Bounding Box(min to max corners): min:(403, 0), max:(479, 175)
Orientation (Theta): 1.27 radians
Eccentricity: 0.92
Perimeter: 59.07
Compactness: 0.62
Component 548:
Area: 307
Centroid: (468.92, 6.40)
Bounding Box(min to max corners): min:(451, 0), max:(479, 19)
Orientation (Theta): 0.40 radians
Eccentricity: 0.82
Perimeter: 63.28
Compactness: 13.05

Total number of components: 23

500:

Component 1:
Area: 1614
Centroid: (7.35, 188.34)
Bounding Box(min to max corners): min:(0, 100), max:(21, 322)
Orientation (Theta): -1.50 radians
Eccentricity: 0.94
Perimeter: 197.11
Compactness: 24.07
Component 48:
Area: 3997
Centroid: (80.32, 424.61)
Bounding Box(min to max corners): min:(34, 319), max:(123, 511)
Orientation (Theta): 1.21 radians
Eccentricity: 0.93
Perimeter: 562.67
Compactness: 79.21
Component 80:
Area: 2847
Centroid: (88.75, 151.81)
Bounding Box(min to max corners): min:(54, 94), max:(120, 254)
Orientation (Theta): -1.51 radians
Eccentricity: 0.89
Perimeter: 91.28
Compactness: 2.93
Component 150:

Computer Vision HW#1

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811296007

Area: 2089
Centroid: (95.38, 345.07)
Bounding Box(min to max corners): min:(79, 272), max:(110, 436)
Orientation (Theta): 1.57 radians
Eccentricity: 0.94
Perimeter: 45.24
Compactness: 0.98

Component 157:
Area: 929
Centroid: (89.20, 206.00)
Bounding Box(min to max corners): min:(81, 120), max:(99, 258)
Orientation (Theta): -1.56 radians
Eccentricity: 0.94
Perimeter: 25.73
Compactness: 0.71

Component 285:
Area: 3632
Centroid: (176.30, 97.39)
Bounding Box(min to max corners): min:(130, 1), max:(205, 173)
Orientation (Theta): -1.31 radians
Eccentricity: 0.84
Perimeter: 33.24
Compactness: 0.30

Component 288:
Area: 2386
Centroid: (173.81, 446.45)
Bounding Box(min to max corners): min:(137, 423), max:(204, 478)
Orientation (Theta): -0.03 radians
Eccentricity: 0.67
Perimeter: 115.18
Compactness: 5.56

Component 304:
Area: 737
Centroid: (176.52, 412.44)
Bounding Box(min to max corners): min:(144, 404), max:(203, 419)
Orientation (Theta): -0.03 radians
Eccentricity: 0.92
Perimeter: 304.40
Compactness: 125.72

Component 345:
Area: 1384
Centroid: (196.74, 281.48)
Bounding Box(min to max corners): min:(183, 239), max:(206, 324)
Orientation (Theta): -1.57 radians
Eccentricity: 0.91
Perimeter: 57.24
Compactness: 2.37

Component 403:
Area: 2065
Centroid: (229.68, 171.12)
Bounding Box(min to max corners): min:(199, 146), max:(255, 214)
Orientation (Theta): -0.61 radians
Eccentricity: 0.73
Perimeter: 494.17
Compactness: 118.26

Component 440:
Area: 1986
Centroid: (267.26, 447.10)
Bounding Box(min to max corners): min:(253, 397), max:(283, 506)
Orientation (Theta): 1.50 radians
Eccentricity: 0.92
Perimeter: 108.56
Compactness: 5.93

Component 453:
Area: 11222
Centroid: (316.58, 377.34)
Bounding Box(min to max corners): min:(280, 282), max:(354, 474)
Orientation (Theta): 1.55 radians
Eccentricity: 0.90

Computer Vision HW#1

Nima Darbandi

811296007

Perimeter: 115.80
Compactness: 1.19

Component 529:

Area: 5655
Centroid: (440.19, 76.64)
Bounding Box(min to max corners): min:(403, 0), max:(479, 175)
Orientation (Theta): 1.27 radians
Eccentricity: 0.92
Perimeter: 59.07
Compactness: 0.62

Total number of components: 13

1000:

Component 1:

Area: 1614
Centroid: (7.35, 188.34)
Bounding Box(min to max corners): min:(0, 100), max:(21, 322)
Orientation (Theta): -1.50 radians
Eccentricity: 0.94
Perimeter: 197.11
Compactness: 24.07

Component 48:

Area: 3997
Centroid: (80.32, 424.61)
Bounding Box(min to max corners): min:(34, 319), max:(123, 511)
Orientation (Theta): 1.21 radians
Eccentricity: 0.93
Perimeter: 562.67
Compactness: 79.21

Component 80:

Area: 2847
Centroid: (88.75, 151.81)
Bounding Box(min to max corners): min:(54, 94), max:(120, 254)
Orientation (Theta): -1.51 radians
Eccentricity: 0.89
Perimeter: 91.28
Compactness: 2.93

Component 150:

Area: 2089
Centroid: (95.38, 345.07)
Bounding Box(min to max corners): min:(79, 272), max:(110, 436)
Orientation (Theta): 1.57 radians
Eccentricity: 0.94
Perimeter: 45.24
Compactness: 0.98

Component 285:

Area: 3632
Centroid: (176.30, 97.39)
Bounding Box(min to max corners): min:(130, 1), max:(205, 173)
Orientation (Theta): -1.31 radians
Eccentricity: 0.84
Perimeter: 33.24
Compactness: 0.30

Component 288:

Area: 2386
Centroid: (173.81, 446.45)
Bounding Box(min to max corners): min:(137, 423), max:(204, 478)
Orientation (Theta): -0.03 radians
Eccentricity: 0.67
Perimeter: 115.18
Compactness: 5.56

Component 345:

Area: 1384
Centroid: (196.74, 281.48)
Bounding Box(min to max corners): min:(183, 239), max:(206, 324)
Orientation (Theta): -1.57 radians
Eccentricity: 0.91

Computer Vision HW#1

Nima Darbandi

811296007

Perimeter: 57.24
Compactness: 2.37

Component 403:

Area: 2065
Centroid: (229.68, 171.12)
Bounding Box(min to max corners): min:(199, 146), max:(255, 214)
Orientation (Theta): -0.61 radians
Eccentricity: 0.73
Perimeter: 494.17
Compactness: 118.26

Component 440:

Area: 1986
Centroid: (267.26, 447.10)
Bounding Box(min to max corners): min:(253, 397), max:(283, 506)
Orientation (Theta): 1.50 radians
Eccentricity: 0.92
Perimeter: 108.56
Compactness: 5.93

Component 453:

Area: 11222
Centroid: (316.58, 377.34)
Bounding Box(min to max corners): min:(280, 282), max:(354, 474)
Orientation (Theta): 1.55 radians
Eccentricity: 0.90
Perimeter: 115.80
Compactness: 1.19

Component 529:

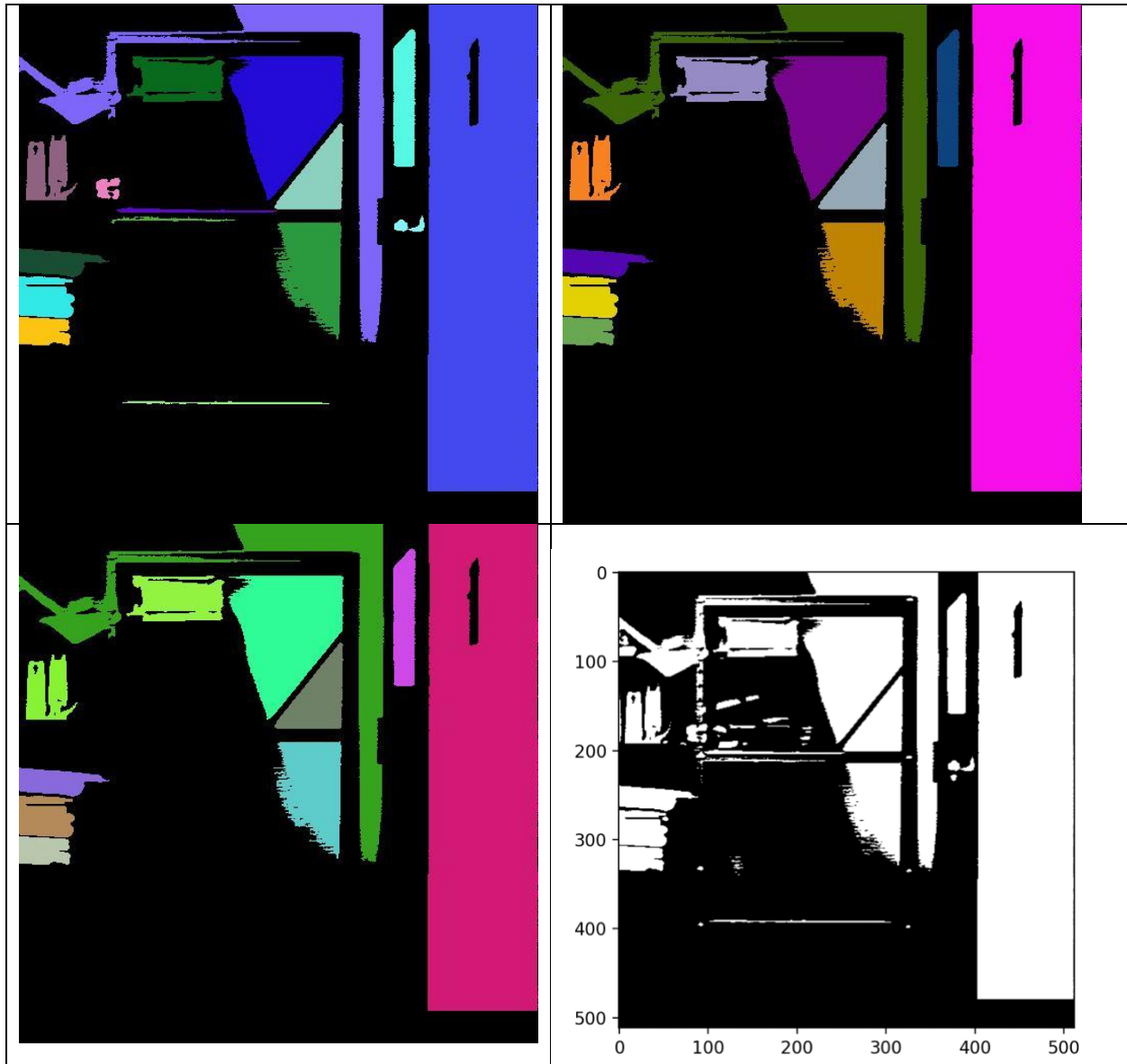
Area: 5655
Centroid: (440.19, 76.64)
Bounding Box(min to max corners): min:(403, 0), max:(479, 175)
Orientation (Theta): 1.27 radians
Eccentricity: 0.92
Perimeter: 59.07
Compactness: 0.62

Total number of components: 11

Computer Vision HW#1

Nima Darbandi

811296007



From top left 200, 500, 1000 pixels min size and 16, 11,, 11 components are detected.
Here is the output data respectively:

200:

Component 1:

Area: 14371

Centroid: (101.58, 259.12)

Bounding Box(min to max corners): min:(0, 0), max:(332, 359)

Orientation (Theta): 1.11 radians

Eccentricity: 0.73

Computer Vision HW#1

Nima Darbandi

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Perimeter: 1279.37
Compactness: 113.90

Component 2:

Area: 50853
Centroid: (242.60, 457.24)
Bounding Box(min to max corners): min:(0, 403), max:(479, 511)
Orientation (Theta): -0.00 radians
Eccentricity: 0.93
Perimeter: 38494.77
Compactness: 29139.83

Component 4:

Area: 2694
Centroid: (94.34, 379.56)
Bounding Box(min to max corners): min:(24, 369), max:(159, 390)
Orientation (Theta): -0.01 radians
Eccentricity: 0.93
Perimeter: 1514.23
Compactness: 851.12

Component 59:

Area: 3329
Centroid: (73.66, 157.42)
Bounding Box(min to max corners): min:(51, 106), max:(95, 202)
Orientation (Theta): 1.54 radians
Eccentricity: 0.84
Perimeter: 682.91
Compactness: 140.09

Component 61:

Area: 10227
Centroid: (103.29, 263.65)
Bounding Box(min to max corners): min:(51, 206), max:(192, 319)
Orientation (Theta): -0.30 radians
Eccentricity: 0.65
Perimeter: 438.68
Compactness: 18.82

Component 171:

Area: 3081
Centroid: (171.88, 294.89)
Bounding Box(min to max corners): min:(117, 252), max:(201, 319)
Orientation (Theta): -0.57 radians
Eccentricity: 0.82
Perimeter: 2381.91
Compactness: 1841.44

Component 192:

Area: 2073
Centroid: (164.08, 27.80)
Bounding Box(min to max corners): min:(129, 7), max:(192, 57)
Orientation (Theta): -0.11 radians
Eccentricity: 0.70
Perimeter: 109.36
Compactness: 5.77

Component 282:

Area: 343
Centroid: (180.76, 86.97)
Bounding Box(min to max corners): min:(171, 76), max:(191, 99)
Orientation (Theta): -1.46 radians
Eccentricity: 0.50
Perimeter: 93.94
Compactness: 25.73

Component 361:

Area: 435
Centroid: (202.95, 162.24)
Bounding Box(min to max corners): min:(200, 96), max:(204, 253)
Orientation (Theta): 1.56 radians
Eccentricity: 0.94
Perimeter: 148.90
Compactness: 50.97

Component 377:

Area: 224
Centroid: (217.98, 384.40)

Computer Vision HW#1

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Bounding Box(min to max corners): min:(208, 370), max:(223, 399)
Orientation (Theta): -1.55 radians
Eccentricity: 0.90
Perimeter: 169.47
Compactness: 128.21
Component 379:
Area: 275
Centroid: (211.45, 145.72)
Bounding Box(min to max corners): min:(209, 91), max:(215, 212)
Orientation (Theta): 1.56 radians
Eccentricity: 0.94
Perimeter: 17.90
Compactness: 1.17
Component 409:
Area: 5243
Centroid: (260.25, 289.83)
Bounding Box(min to max corners): min:(214, 251), max:(329, 317)
Orientation (Theta): 0.21 radians
Eccentricity: 0.81
Perimeter: 132.80
Compactness: 3.36
Component 438:
Area: 1619
Centroid: (253.60, 34.95)
Bounding Box(min to max corners): min:(240, 0), max:(268, 88)
Orientation (Theta): 1.55 radians
Eccentricity: 0.91
Perimeter: 95.38
Compactness: 5.62
Component 458:
Area: 1914
Centroid: (288.76, 25.10)
Bounding Box(min to max corners): min:(268, 0), max:(308, 54)
Orientation (Theta): 1.37 radians
Eccentricity: 0.68
Perimeter: 326.50
Compactness: 55.70
Component 492:
Area: 1269
Centroid: (321.72, 23.54)
Bounding Box(min to max corners): min:(308, 0), max:(335, 50)
Orientation (Theta): 1.53 radians
Eccentricity: 0.83
Perimeter: 930.58
Compactness: 682.40
Component 540:
Area: 389
Centroid: (392.27, 203.72)
Bounding Box(min to max corners): min:(391, 102), max:(393, 305)
Orientation (Theta): 1.57 radians
Eccentricity: 0.94
Perimeter: 200.49
Compactness: 103.33

Total number of components: 16

500:

Component 1:
Area: 14371
Centroid: (101.58, 259.12)
Bounding Box(min to max corners): min:(0, 0), max:(332, 359)
Orientation (Theta): 1.11 radians
Eccentricity: 0.73
Perimeter: 1279.37
Compactness: 113.90
Component 2:
Area: 50853
Centroid: (242.60, 457.24)
Bounding Box(min to max corners): min:(0, 403), max:(479, 511)

Computer Vision HW#1

Nima Darbandi

811296007

Orientation (Theta): -0.00 radians

Eccentricity: 0.93

Perimeter: 38494.77

Compactness: 29139.83

Component 4:

Area: 2694

Centroid: (94.34, 379.56)

Bounding Box(min to max corners): min:(24, 369), max:(159, 390)

Orientation (Theta): -0.01 radians

Eccentricity: 0.93

Perimeter: 1514.23

Compactness: 851.12

Component 59:

Area: 3329

Centroid: (73.66, 157.42)

Bounding Box(min to max corners): min:(51, 106), max:(95, 202)

Orientation (Theta): 1.54 radians

Eccentricity: 0.84

Perimeter: 682.91

Compactness: 140.09

Component 61:

Area: 10227

Centroid: (103.29, 263.65)

Bounding Box(min to max corners): min:(51, 206), max:(192, 319)

Orientation (Theta): -0.30 radians

Eccentricity: 0.65

Perimeter: 438.68

Compactness: 18.82

Component 171:

Area: 3081

Centroid: (171.88, 294.89)

Bounding Box(min to max corners): min:(117, 252), max:(201, 319)

Orientation (Theta): -0.57 radians

Eccentricity: 0.82

Perimeter: 2381.91

Compactness: 1841.44

Component 192:

Area: 2073

Centroid: (164.08, 27.80)

Bounding Box(min to max corners): min:(129, 7), max:(192, 57)

Orientation (Theta): -0.11 radians

Eccentricity: 0.70

Perimeter: 109.36

Compactness: 5.77

Component 409:

Area: 5243

Centroid: (260.25, 289.83)

Bounding Box(min to max corners): min:(214, 251), max:(329, 317)

Orientation (Theta): 0.21 radians

Eccentricity: 0.81

Perimeter: 132.80

Compactness: 3.36

Component 438:

Area: 1619

Centroid: (253.60, 34.95)

Bounding Box(min to max corners): min:(240, 0), max:(268, 88)

Orientation (Theta): 1.55 radians

Eccentricity: 0.91

Perimeter: 95.38

Compactness: 5.62

Component 458:

Area: 1914

Centroid: (288.76, 25.10)

Bounding Box(min to max corners): min:(268, 0), max:(308, 54)

Orientation (Theta): 1.37 radians

Eccentricity: 0.68

Perimeter: 326.50

Compactness: 55.70

Component 492:

Computer Vision HW#1

Nima Darbandi

811296007

Area: 1269
Centroid: (321.72, 23.54)
Bounding Box(min to max corners): min:(308, 0), max:(335, 50)
Orientation (Theta): 1.53 radians
Eccentricity: 0.83
Perimeter: 930.58
Compactness: 682.40

Total number of components: 11

1000:

Component 1:

Area: 14371
Centroid: (101.58, 259.12)
Bounding Box(min to max corners): min:(0, 0), max:(332, 359)
Orientation (Theta): 1.11 radians
Eccentricity: 0.73
Perimeter: 1279.37
Compactness: 113.90

Component 2:

Area: 50853
Centroid: (242.60, 457.24)
Bounding Box(min to max corners): min:(0, 403), max:(479, 511)
Orientation (Theta): -0.00 radians
Eccentricity: 0.93
Perimeter: 38494.77
Compactness: 29139.83

Component 4:

Area: 2694
Centroid: (94.34, 379.56)
Bounding Box(min to max corners): min:(24, 369), max:(159, 390)
Orientation (Theta): -0.01 radians
Eccentricity: 0.93
Perimeter: 1514.23
Compactness: 851.12

Component 59:

Area: 3329
Centroid: (73.66, 157.42)
Bounding Box(min to max corners): min:(51, 106), max:(95, 202)
Orientation (Theta): 1.54 radians
Eccentricity: 0.84
Perimeter: 682.91
Compactness: 140.09

Component 61:

Area: 10227
Centroid: (103.29, 263.65)
Bounding Box(min to max corners): min:(51, 206), max:(192, 319)
Orientation (Theta): -0.30 radians
Eccentricity: 0.65
Perimeter: 438.68
Compactness: 18.82

Component 171:

Area: 3081
Centroid: (171.88, 294.89)
Bounding Box(min to max corners): min:(117, 252), max:(201, 319)
Orientation (Theta): -0.57 radians
Eccentricity: 0.82
Perimeter: 2381.91
Compactness: 1841.44

Component 192:

Area: 2073
Centroid: (164.08, 27.80)
Bounding Box(min to max corners): min:(129, 7), max:(192, 57)
Orientation (Theta): -0.11 radians
Eccentricity: 0.70
Perimeter: 109.36
Compactness: 5.77

Component 409:

Computer Vision HW#1

Nima Darbandi

811296007

Area: 5243

Centroid: (260.25, 289.83)

Bounding Box(min to max corners): min:(214, 251), max:(329, 317)

Orientation (Theta): 0.21 radians

Eccentricity: 0.81

Perimeter: 132.80

Compactness: 3.36

Component 438:

Area: 1619

Centroid: (253.60, 34.95)

Bounding Box(min to max corners): min:(240, 0), max:(268, 88)

Orientation (Theta): 1.55 radians

Eccentricity: 0.91

Perimeter: 95.38

Compactness: 5.62

Component 458:

Area: 1914

Centroid: (288.76, 25.10)

Bounding Box(min to max corners): min:(268, 0), max:(308, 54)

Orientation (Theta): 1.37 radians

Eccentricity: 0.68

Perimeter: 326.50

Compactness: 55.70

Component 492:

Area: 1269

Centroid: (321.72, 23.54)

Bounding Box(min to max corners): min:(308, 0), max:(335, 50)

Orientation (Theta): 1.53 radians

Eccentricity: 0.83

Perimeter: 930.58

Compactness: 682.40

Total number of components: 11